Question and delivered model

Predict correctness for each contact in an actual MarinFold rollout, allowing later contacts to revise earlier judgments. Support a single contact and variable-length prefixes. Also estimate current precision and recall.

Released checkpoint: exp356-exp277-frozen-contact-pdb50k-v2/step-18000. Training and reserved-test evaluation are complete. Public model bytes and shipped inference were independently verified.

Architecture

Frozen exp277 step-266344 causal Qwen3 features feed a four-layer, 512-wide bidirectional contact encoder. Each contact contributes all three token states. Global features include the sequence-prompt boundary, last supplied token and prefix size.

22.18 million trainable parameters; 1.488 billion total. Per-contact sigmoid probabilities, mean unique-pair precision and a pooled recall head. Inputs are cropped before encoding; future tokens and labels are absent.

Experimental data

50,000 training chain structures, 46,933 PDB entries, 34,894 distinct supplied sequences and 9,074 connected related groups. Validation: 1,552 structures / 287 groups. Reserved test: 1,473 structures / 295 groups.

All 32 generation shards completed. Audit retained 423,371 real rollouts, including 399,213 training trajectories and 109.3 million contact labels across all splits. Reference completeness and token/label alignment were checked. Every target retains trajectories.

Training and checkpoint selection

Eight CoreWeave H100s, three epochs, 18,714 optimizer steps, global batch 64. Prefix sampler: 25% full, 25% short, 50% uniform. Frozen features beat full-backbone bidirectional fine-tuning at matched step 2,000 and ran about four times faster.

Validation selected step 18000 from the predeclared candidate pool; its eight-prefix loss is 0.431166. The choice was frozen before evaluating the reserved test. Storage recovery preserved exact model, optimizer and schedule state.

Reserved-test quality

All 1,473 test structures / 295 groups. Full-rollout AUROC 0.7933; Brier 0.1438. Precision MAE 0.0521; recall MAE 0.0528. One-contact Brier 0.1350.

At 64 supplied contacts, first-contact Brier improves by 0.0067 (95% group-bootstrap interval 0.0047–0.0088), excluding end markers. Keeping the top-scored half yields precision 0.5788, versus 0.4019 for emission order.

Downstream use and limits

Selecting a rollout by predicted precision gives 0.4504, versus 0.4033 for random selection. Weighted aggregation R-precision 0.5135, versus 0.4987 for frequency alone.

Measured scope: exp277 single-chain rollouts, 32–1,000 residues, temperature 1, top-p 0.95. Proofreader splits are held out; generator-pretraining exclusion is not established. Other generators, complexes and edited rollouts remain unvalidated.

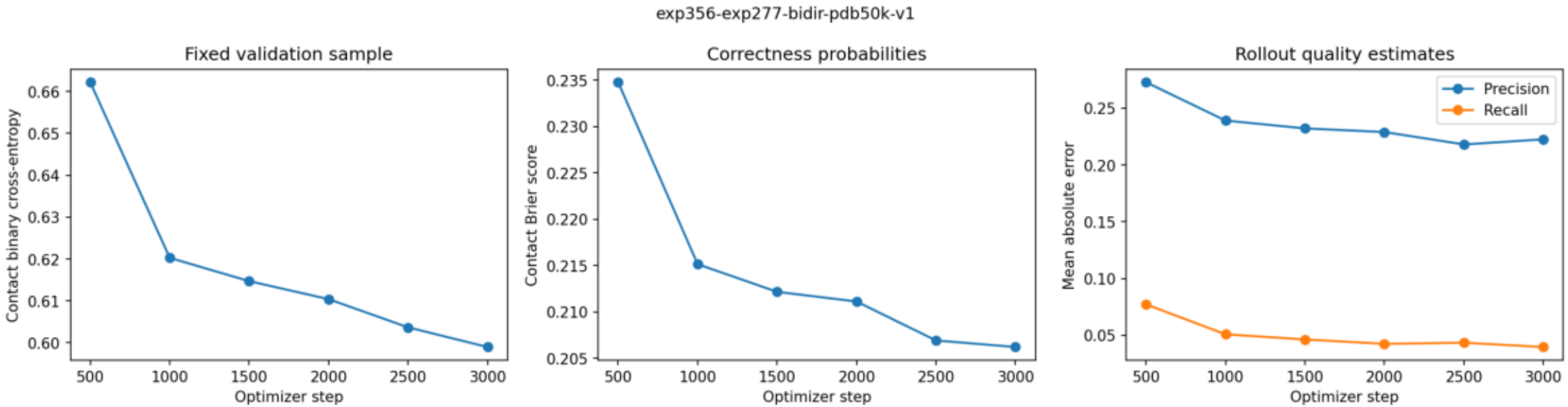
Reproducibility and release checks

Public checkpoint includes tokenizer, code, pinned dependencies and hashes. Raw rollouts, targets, split provenance, per-rollout tables, plots and predictor timings are public.

Twenty-one behavioral tests pass. Optimizer recovery was exercised on one and eight GPUs. Independent anonymous download verified every checkpoint hash and ran shipped inference offline for one contact and a complete rollout.

exp356-exp277-bidir-pdb50k-v1_learning.png

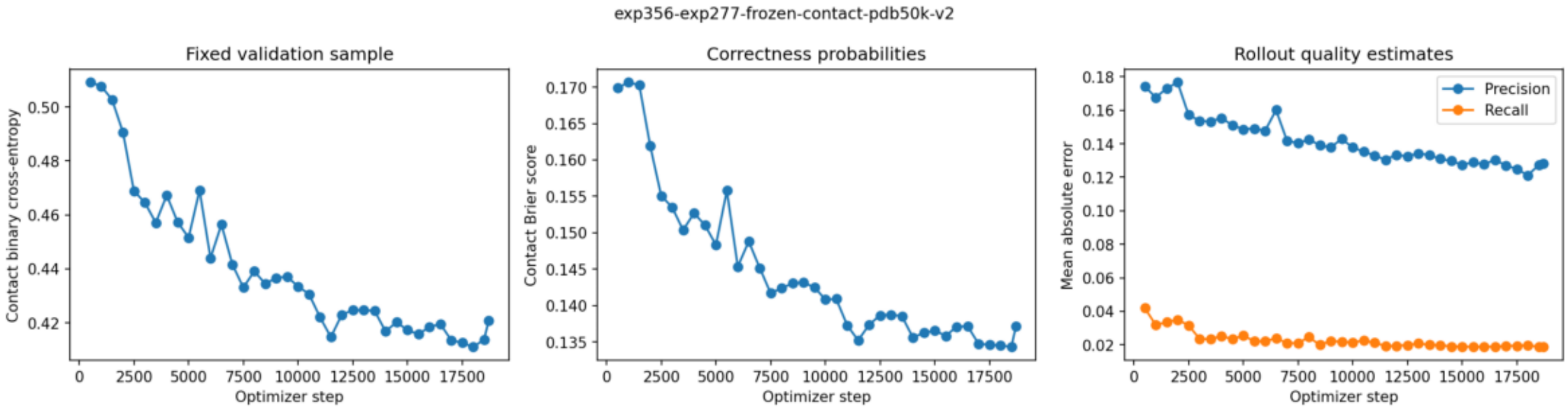
Fixed 256-rollout validation sample at prefixes 1, 4, 16 and full. These training diagnostics are separate from the complete held-out evaluation.




```
$ plot_learning.py --run exp356-exp277-bidir-pdb50k-v1
```

exp356-exp277-frozen-contact-pdb50k-v2_learning.png

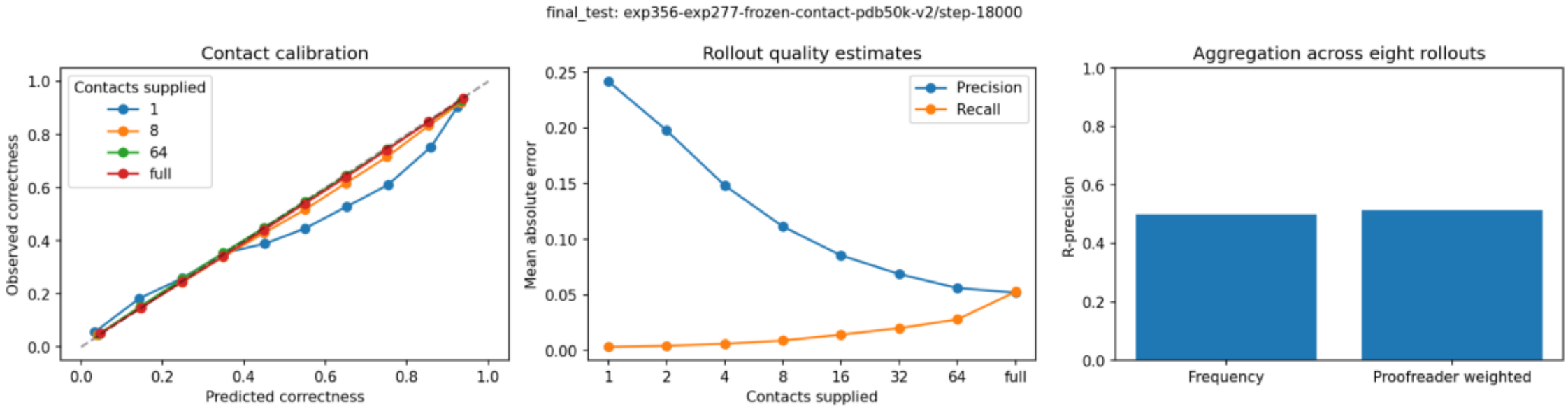
Fixed 256-rollout validation sample at prefixes 1, 4, 16 and full. These training diagnostics are separate from the complete held-out evaluation.




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$ plot_learning.py --run exp356-exp277-frozen-contact-pdb50k-v2
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final_test_quality.png

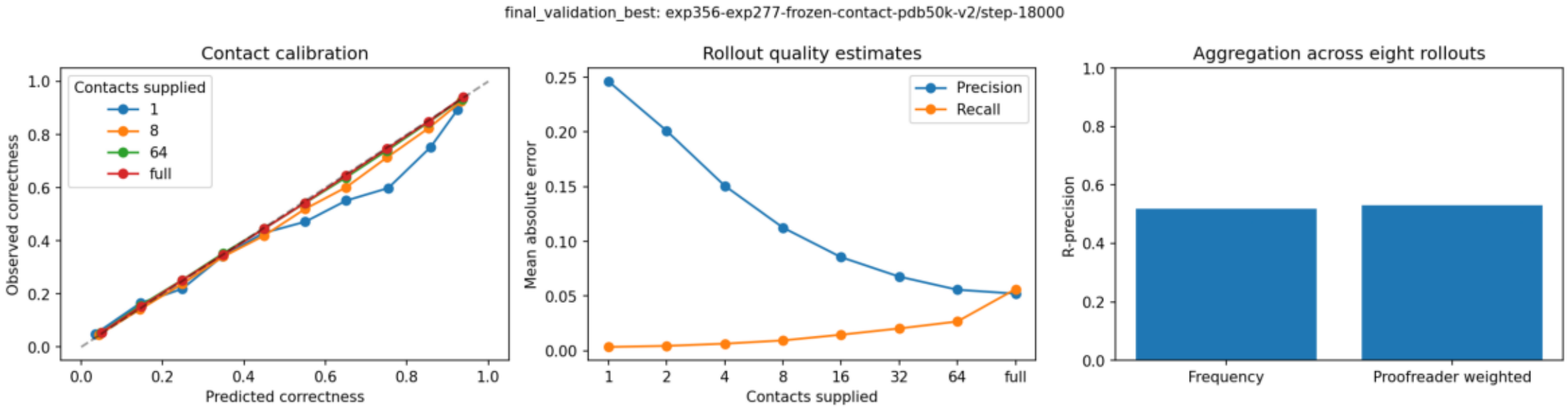
final_test, exp356-exp277-frozen-contact-pdb50k-v2/step-18000. Protein means; calibration bins pool contacts. Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label final_test --replot-only
```

final_validation_best_quality.png

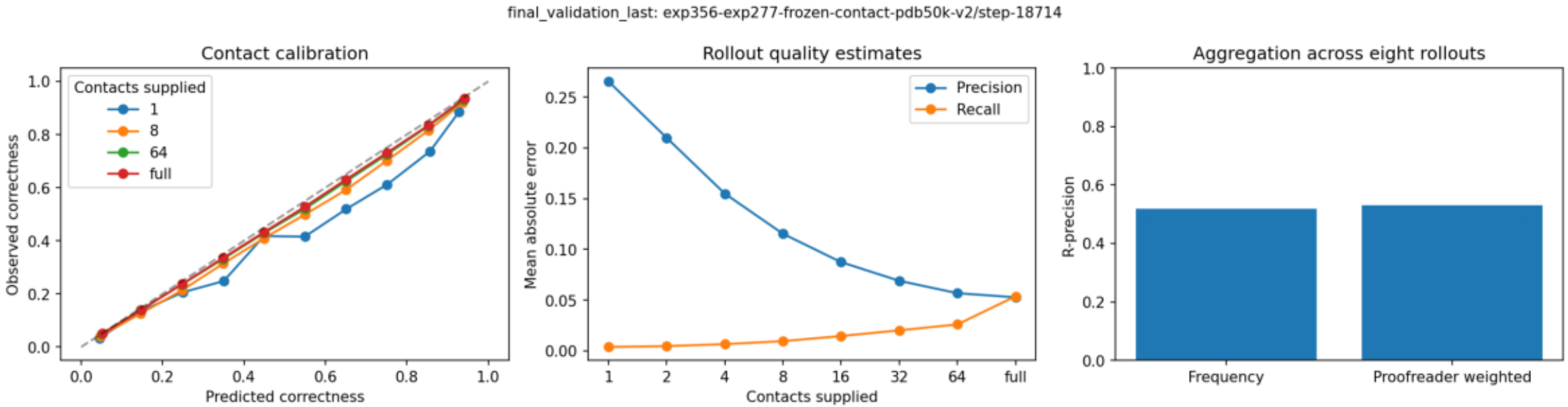
final_validation_best, exp356-exp277-frozen-contact-pdb50k-v2/step-18000. Protein means; calibration bins pool contacts.
Confidence intervals in metrics JSON resample related groups.




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$ analyze.py --label final_validation_best --replot-only
```

final_validation_last_quality.png

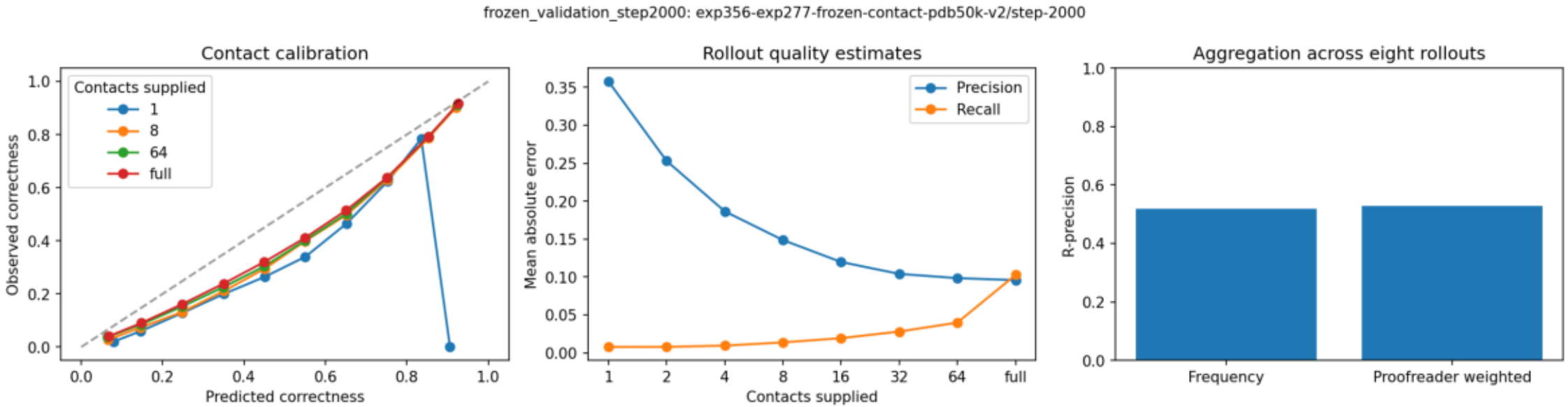
final_validation_last, exp356-exp277-frozen-contact-pdb50k-v2/step-18714. Protein means; calibration bins pool contacts.
Confidence intervals in metrics JSON resample related groups.




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$ analyze.py --label final_validation_last --replot-only
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frozen_validation_step2000_quality.png

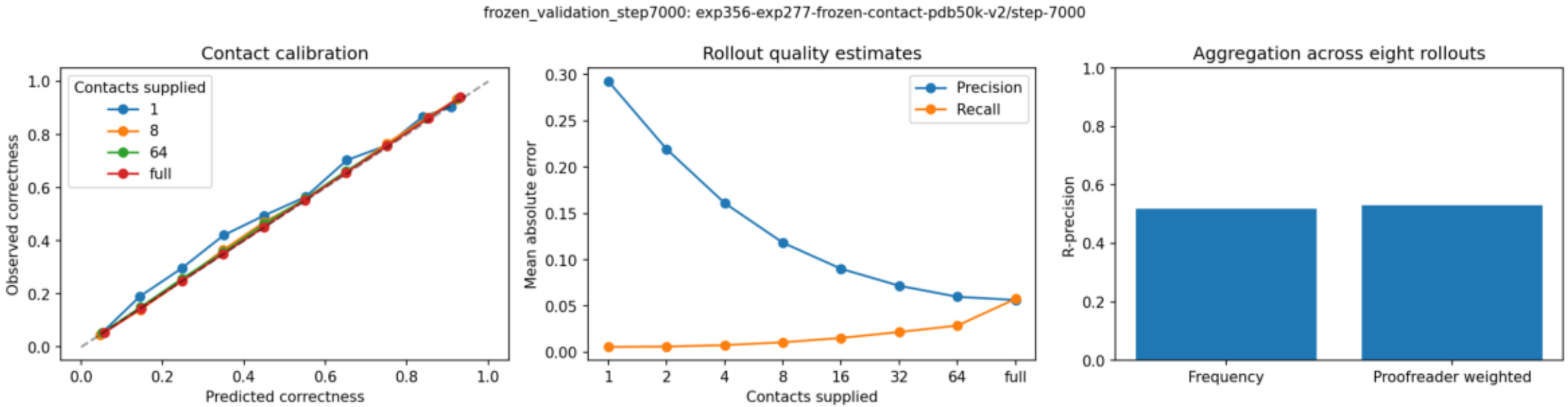
frozen_validation_step2000, exp356-exp277-frozen-contact-pdb50k-v2/step-2000. Protein means; calibration bins pool contacts.
Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label frozen_validation_step2000 --replot-only
```

frozen_validation_step7000_quality.png

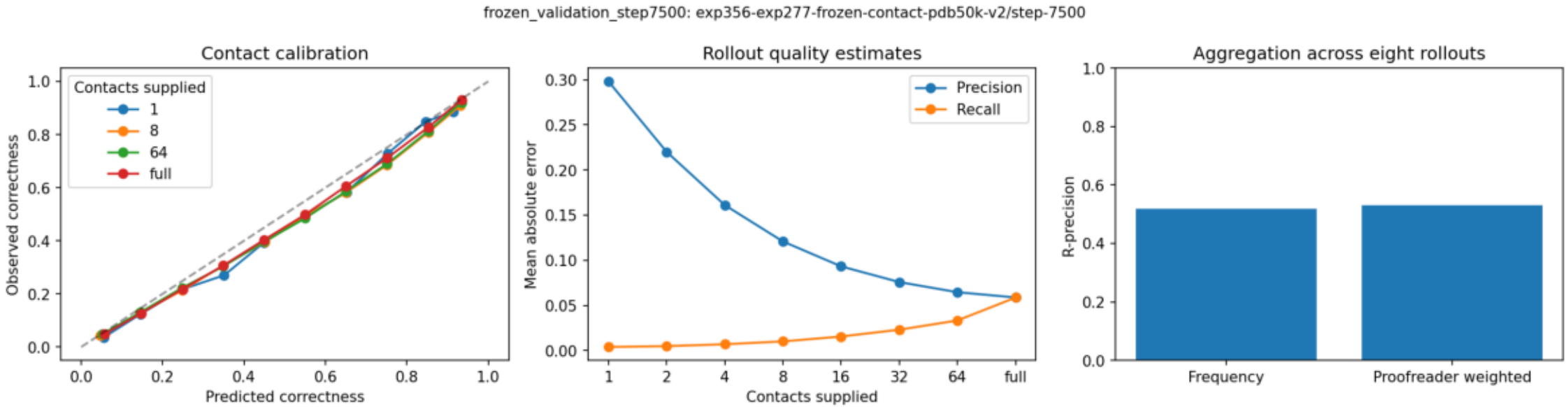
frozen_validation_step7000, exp356-exp277-frozen-contact-pdb50k-v2/step-7000. Protein means; calibration bins pool contacts.
Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label frozen_validation_step7000 --replot-only
```

frozen_validation_step7500_quality.png

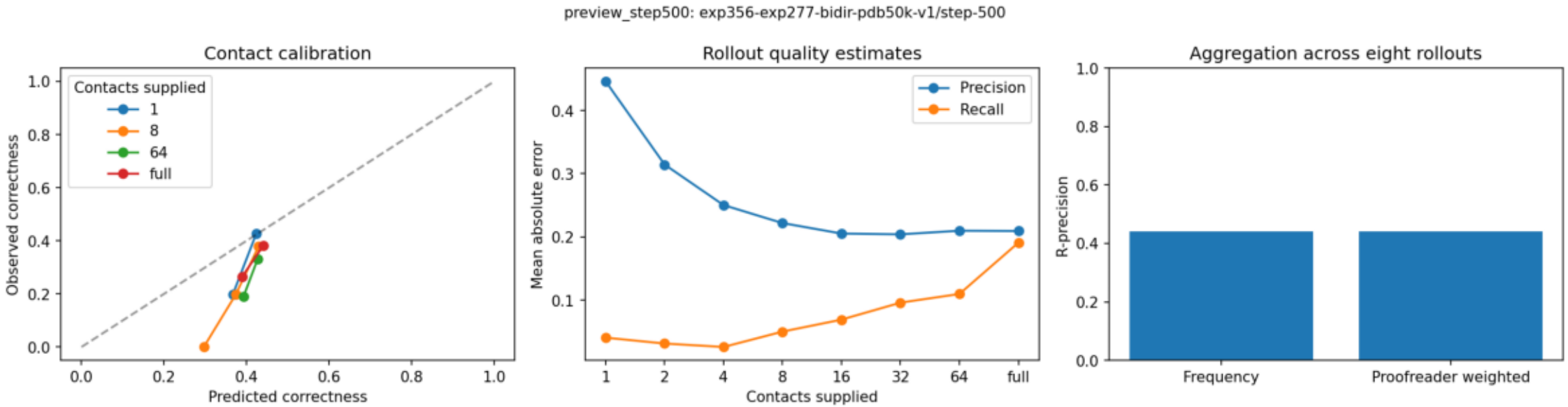
frozen_validation_step7500, exp356-exp277-frozen-contact-pdb50k-v2/step-7500. Protein means; calibration bins pool contacts.
Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label frozen_validation_step7500 --replot-only
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preview_step500_quality.png

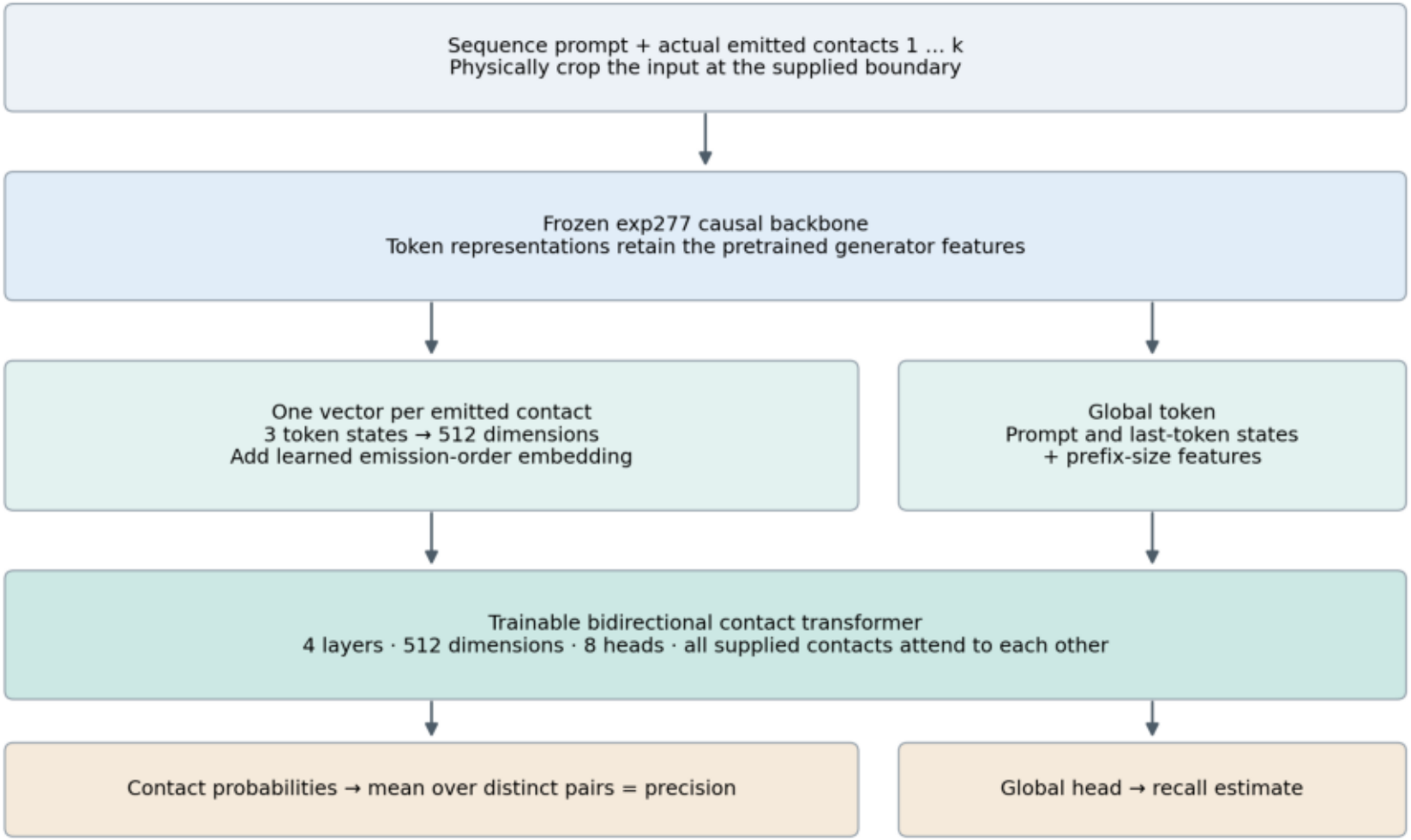
preview_step500, exp356-exp277-bidir-pdb50k-v1/step-500. Protein means; calibration bins pool contacts. Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label preview_step500 --replot-only
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report_architecture.png

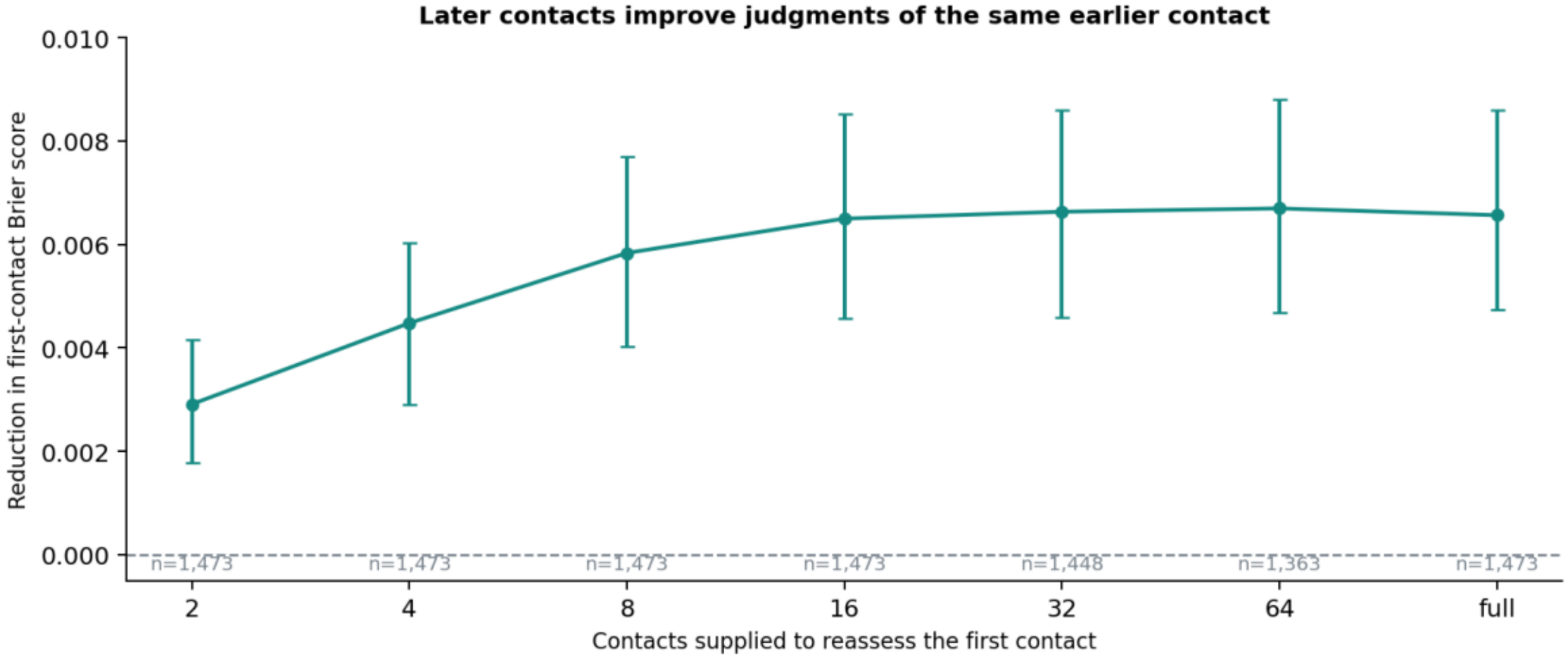
Figure 1. The frozen causal backbone feeds a trainable bidirectional contact encoder. Later supplied contacts can revise earlier scores; unsupplied continuation is absent. There are 22,181,378 trainable parameters out of 1,487,731,202 total.



$ plot_report.py

report_context.png

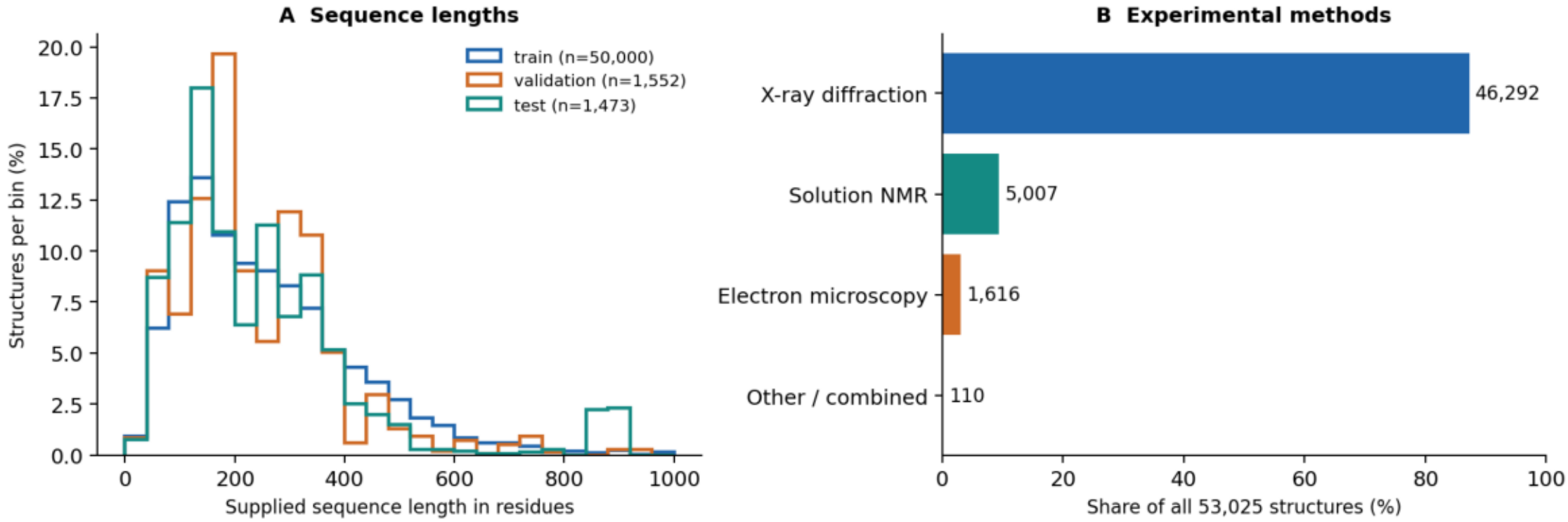
Figure 5. Paired first-contact Brier reduction relative to supplying only the first contact. Positive values mean improvement. Each comparison uses the same eligible trajectories in both conditions; n denotes structures. Bars are 95% group-bootstrap intervals. Explicit fixed prefixes exclude the later end marker; full includes a genuine end marker when present.



$ plot_report.py

report_dataset.png

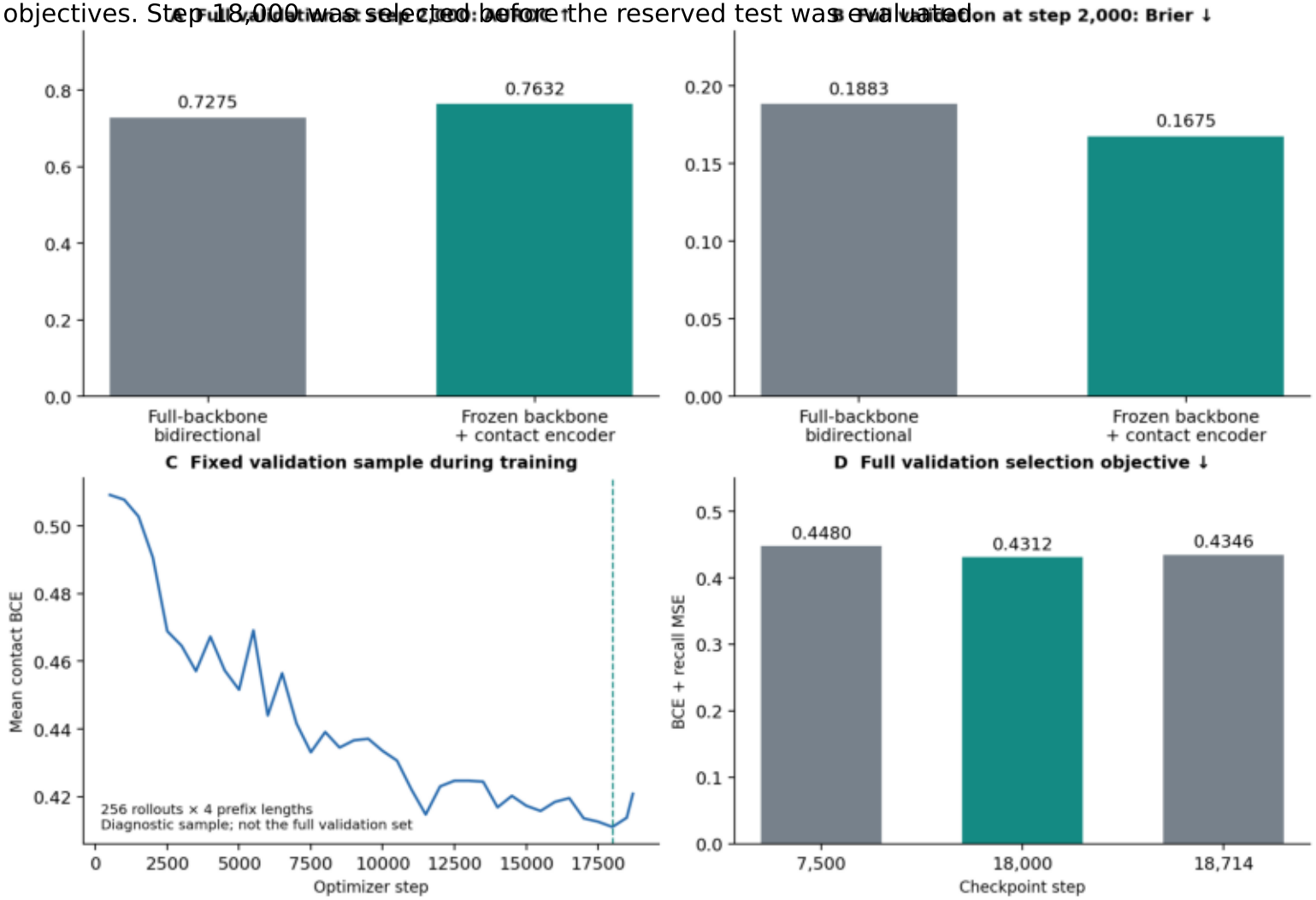
Figure 2. Supplied sequence lengths and experimental methods in the 53,025-structure corpus. Length histograms are normalized separately within each split; method counts combine all splits. These are chain structures, not independent sequences.



$ plot_report.py

report_development.png

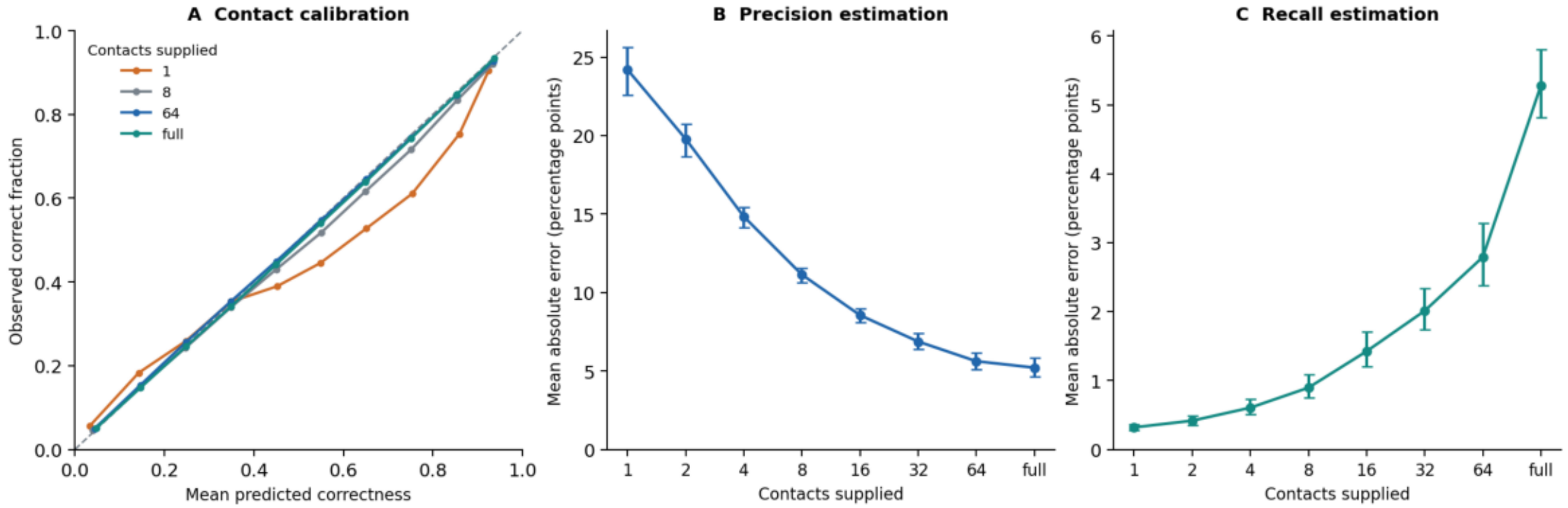
Figure 3. Development evidence. A–B: matched-step architecture comparison over all validation structures. C: fixed 256-rollout diagnostic sample, averaging prefixes 1, 4, 16 and full. D: final selection on all validation structures, averaging eight prefix-specific BCE + recall-MSE objectives. Step 18,000 was selected before the reserved test was evaluated.




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$ plot_report.py
```

report_test_quality.png

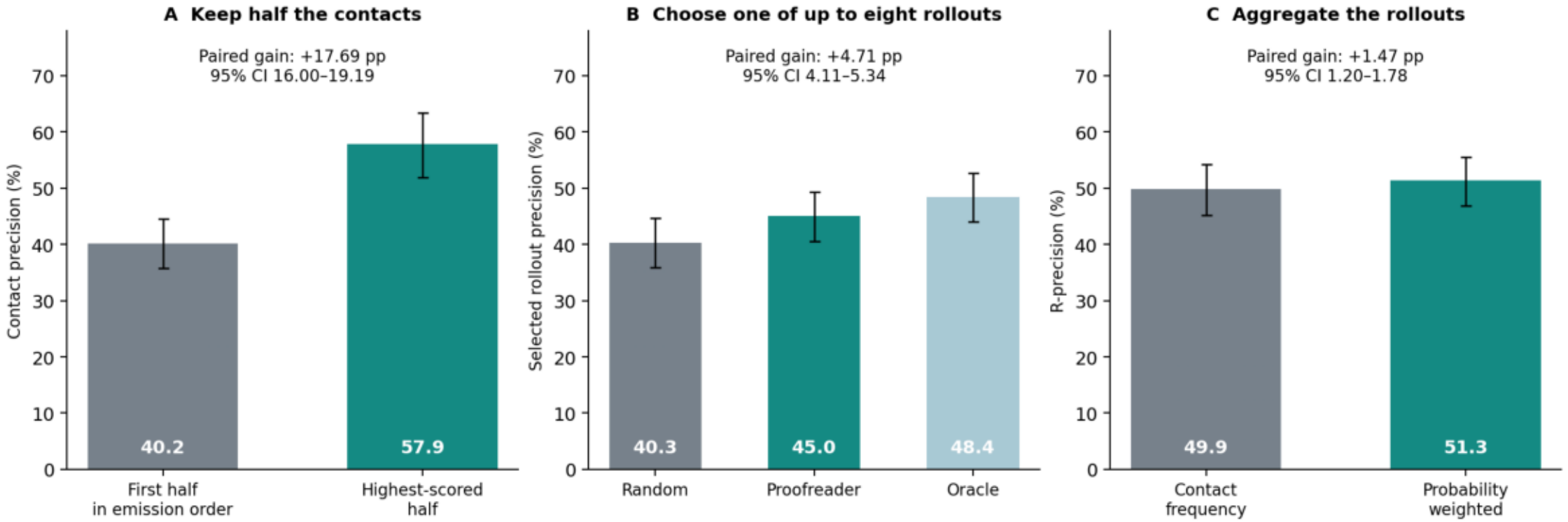
Figure 4. Reserved-test quality at the selected step-18,000 checkpoint. Calibration pools contacts within probability bins. MAE first averages rollouts within each structure and then structures; bars are saved 95% intervals from 1,000 related-group bootstrap replicates. Longer fixed prefixes include only sufficiently long rollouts. Full is a separate variable-length condition.



$ plot_report.py

report_utility.png

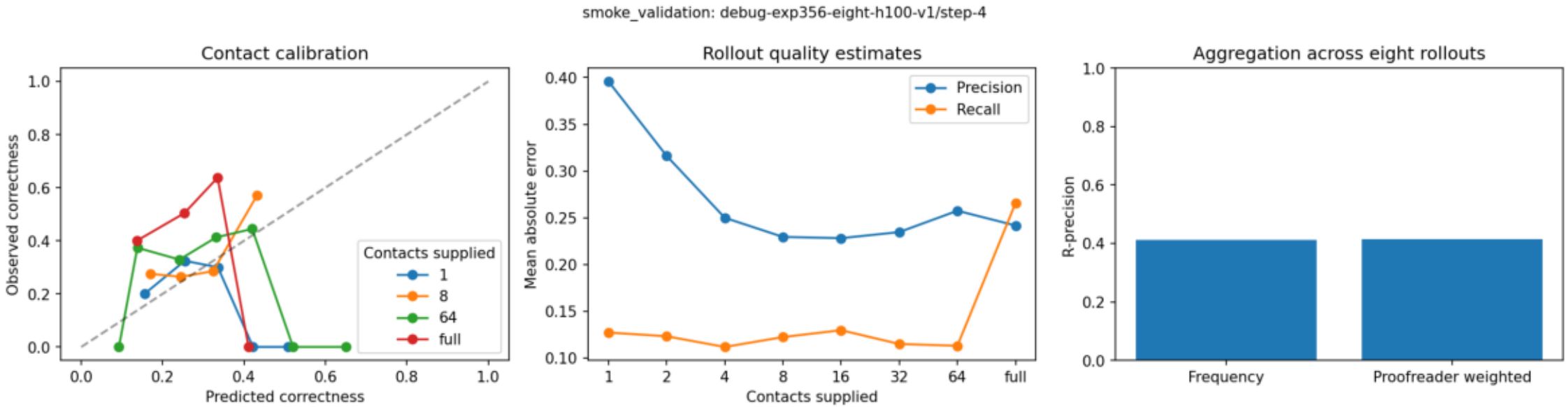
Figure 6. Reserved-test downstream uses of the released scores. A compares equal contact budgets within each rollout; B selects a whole rollout by predicted precision, with an unattainable oracle for context; C ranks pooled distinct pairs by occurrence count or summed correctness probability. Bars show structure-mean scores and 95% group-bootstrap intervals. Paired gains and their intervals compare the proofreader with the first baseline in each panel; pp means percentage points.



$ plot_report.py

smoke_validation_quality.png

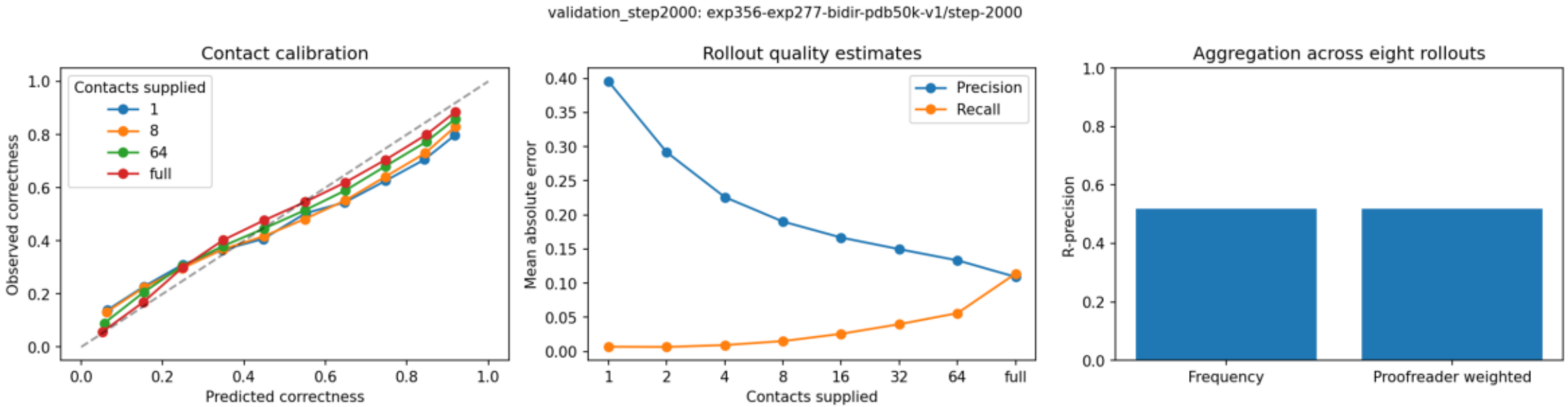
smoke_validation, debug-exp356-eight-h100-v1/step-4. Protein means; calibration bins pool contacts. Confidence intervals in metrics JSON resample related groups.




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$ analyze.py --label smoke_validation --replot-only
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validation_step2000_quality.png

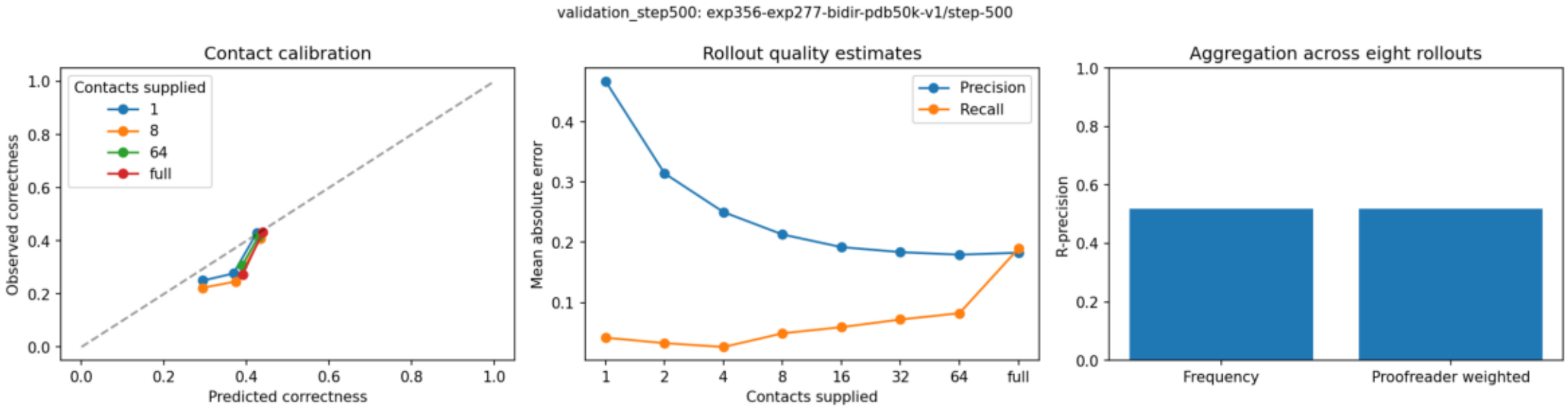
validation_step2000, exp356-exp277-bidir-pdb50k-v1/step-2000. Protein means; calibration bins pool contacts. Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label validation_step2000 --replot-only
```

validation_step500_quality.png

validation_step500, exp356-exp277-bidir-pdb50k-v1/step-500. Protein means; calibration bins pool contacts. Confidence intervals in metrics JSON resample related groups.




```
$ analyze.py --label validation_step500 --replot-only
```